

Statistics 6910, Section 003, Spring 2013, Quiz 2 (25 Points)

Friday, January 25, 2013

Your Name: _____

Question 1: Matrices (8 Points)

The matrix `mat` is created as follows in R:

```
mat = matrix(1:8, nrow = 4, ncol = 2, byrow = TRUE)
rownames(mat) = letters[1:4]
colnames(mat) = letters[1:2]
mat
> mat
  a b
a 1 2
b 3 4
c 5 6
d 7 8
```

For each of the following 8 blocks of R expressions, indicate the resulting output. Note that some of the expressions change the content of `mat` while others leave it unchanged. You should assume that these expressions are **evaluated sequentially**. So, if `mat` was modified via a previous expression, you need to proceed with the modified content of `mat`, and not with the original content. Read carefully!

```
#1
mat[, "a"] = 10
mat
```

```
#2
mat[-(2:3), 2] = 20
mat
```

```
#3
t(mat)
```

```
#4  
mat + 5
```

```
#5  
mat[c(TRUE, TRUE, FALSE, FALSE), c(FALSE, TRUE)]
```

```
#6  
mat[4, ] = mat[1, ] + mat[2, ] + mat[3, ]  
mat
```

```
#7  
mat < 20
```

```
#8  
mat[, ] = 40  
mat
```

Question 2: Sequences (6 Points)

Use `rep`, `seq`, or `:` for the following tasks. Be as efficient as possible.

- Create a vector of length 100, the first 50 values are 1 and the last 50 are 2.

- Create a vector with values:

```
[1] -3.1 -2.1 -1.1 -0.1 0.9
```

- Create a vector with values:

```
[1] 0 0 0 0 0 1 1 1 1 2 2 2 3 3 4
```

Question 3: Earthquakes (11 Points)

When dealing with earthquakes, we are usually interested in their locations, and not only in their magnitudes. In this question, we will focus on earthquakes that occurred in the San Francisco area.

Below are a few expressions and results that should help to remind you on the structure of the earthquake data:

```
CAquakes = read.table(file = "http://www.consrv.ca.gov/cgs/rghm/quakes/Documents/ms49epicenters.txt",
                      header = TRUE)

> dim(CAquakes)
[1] 383  4
>
> class(CAquakes)
[1] "data.frame"
>
> class(CAquakes$Latitude)
[1] "numeric"
>
> class(CAquakes$Longitude)
[1] "numeric"
>
> head(CAquakes)
      Date Latitude Longitude  M
1 18001011    36.8   -121.5 5.5
2 18001122    32.9   -117.8 6.3
3 18030000    34.2   -118.1 5.5
4 18030525    32.8   -117.1 5.5
5 18060325    34.4   -119.7 5.5
6 18080621    37.8   -122.6 5.5
>
> tail(CAquakes)
      Date Latitude Longitude  M
378 19951228  38.7210 -119.6590 5.5
379 19970122  40.2720 -124.3940 5.6
380 19981127  40.6670 -125.3840 5.6
381 19991016  34.6815 -116.2848 5.8
382 19991016  34.6000 -116.2700 7.1
383 19991016  34.4400 -116.2500 5.7
```

The map (see Figure 1) has been created in R with the following commands (for now, just ignore these commands — but feel free to experiment with them at home):

```
library(RgoogleMaps)

bb = qbbox(c(min(CAQuakes$Latitude), max(CAQuakes$Latitude)), c(min(CAQuakes$Longitude), max(CAQuakes$Longitude)))

MyMap = GetMap.bbox(bb$lonR, bb$latR, destfile = "CAQuakes.png", GRAYSCALE = TRUE)

tmp = PlotOnStaticMap(MyMap, lat = CAQuakes$Latitude, lon = CAQuakes$Longitude,
  cex = 1.5, pch = 20, col = "red", add = FALSE, axes = TRUE, mar = c(3, 3, 0, 0))

tmp = PlotOnStaticMap(MyMap, lat = c(36, 36, 39, 36), lon = c(-124, -120, -120, -124),
  col = "black", add = TRUE, FUN = lines, lwd = 2)
```

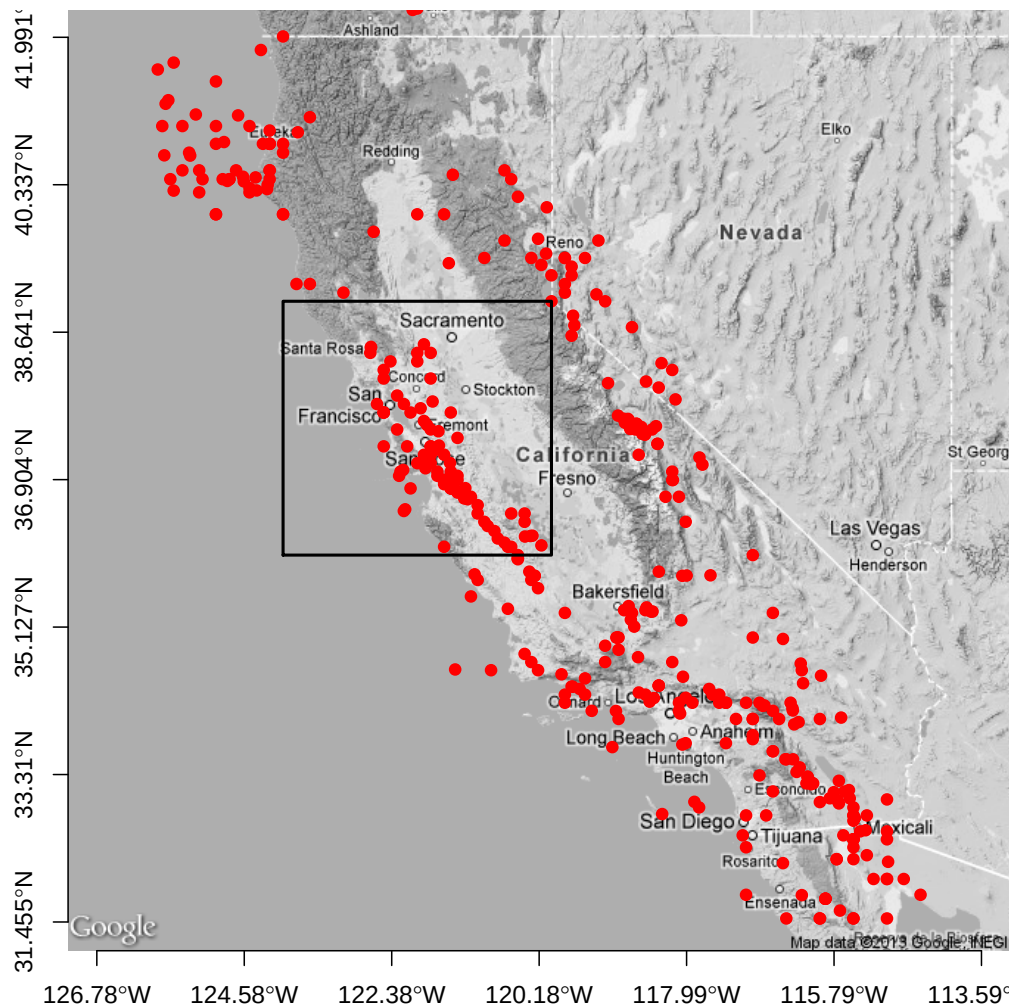


Figure 1: Location of earthquakes in California from 1800 to 1999. The area of interest around San Francisco is inside the rectangular area.

For this question, assume that we are primarily interested in earthquakes that occurred in the San Francisco area. This is the rectangular area (strictly) inside the black box in Figure 1. The lower left corner for this area is (Latitude, Longitude) = (36, -124) and the upper right corner is (Latitude, Longitude) = (39, -120). Recall that negative values for longitude represent locations west of the Greenwich Meridian, i.e., a longitude of -120 alternatively can be written as 120°W.

First, create a logical vector (of length 383), called `inSFArea`, that contains the information whether the epicenter of an earthquake was (strictly) inside the area specified above, i.e., it is `TRUE` if (Latitude, Longitude) is (strictly) inside this area and `FALSE` otherwise. Be careful with the signs and the directions of inequalities! Use this vector as part of your answer in all following parts.

`inSFArea =`

Now, indicate R expressions to answer the following 4 questions. As usual, be as efficient as possible.

1. How many earthquakes were in the area of interest?
2. What was the average (= mean) magnitude of the earthquakes in the area of interest?
3. Report the dates of the earthquakes in the area of interest since January 1, 1990.
4. What was the highest magnitude of an earthquake in the area of interest from January 1, 1900, to December 31, 1910 — and when did it occur? You may want to use more than one expression here (but you still need to be as efficient as possible!).